WORKED SOLUTION

# STEP II 2010, Question 6

Pure • Regular Tetrahedron Geometry

**STEP II 2010, Question 6**

Read the question before you go on. It is in the STEP Support Programme's question database:

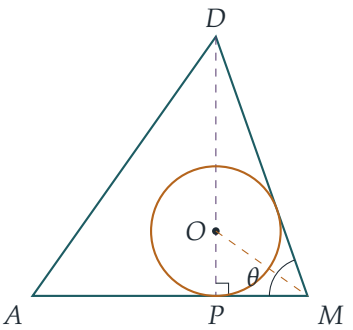
step.maths.org/questions/10-s2-q6

## The key idea

A regular tetrahedron is so symmetric that almost everything happens in one vertical cross-section: the plane through an edge-midpoint, the opposite vertex and the apex. In that plane every 3D question becomes a right-angled triangle. For the sphere, symmetry tells us where the centre must be, and the angle from part (ii) tells us how far it sits from each face.

## Solution

Throughout, let $M$ be the midpoint of $BC$. Every face is an equilateral triangle of side 1, so each median (such as $AM$ or $DM$) has length $\sqrt{1-\frac{1}{4}} = \frac{\sqrt{3}}{2}$.




**Part (i)**

Since $D$ is equidistant from $A$, $B$ and $C$, and $DP$ is perpendicular to the plane $ABC$, Pythagoras gives $PA^2 = DA^2 - DP^2 = DB^2 - DP^2 = PB^2$, and likewise $PB = PC$. So $P$ is the circumcentre of the equilateral triangle $ABC$, which is also its centroid. The

centroid lies two-thirds of the way along each median, so

$$PA = \tfrac{2}{3} \cdot \tfrac{\sqrt{3}}{2} = \tfrac{1}{\sqrt{3}}, \qquad PM = \tfrac{1}{3} \cdot \tfrac{\sqrt{3}}{2} = \tfrac{1}{2\sqrt{3}}.$$

Then, from the right angle at $P$ in triangle $DPA$,

$$PD^2 = DA^2 - PA^2 = 1 - \tfrac{1}{3} = \tfrac{2}{3}, \qquad \text{so} \qquad PD = \sqrt{\tfrac{2}{3}} = \frac{\sqrt{6}}{3}.$$

**Part (ii)**

All six edges look the same (any edge can be moved onto any other by a symmetry of the tetrahedron), so it is enough to find the angle $\theta$ between faces $ABC$ and $DBC$, which meet along $BC$.

**Ask yourself** Which angle in the diagram actually *is* the angle between the two faces?

The angle between two planes meeting in a line is measured using two lines, one in each plane, both perpendicular to the common line at the same point. In the equilateral triangle $ABC$ the median $AM$ is perpendicular to $BC$; in the equilateral triangle $DBC$ the median $DM$ is perpendicular to $BC$. So $\theta = \angle AMD$, and since $P$ lies on $AM$ between $A$ and $M$, also $\theta = \angle PMD$.

Triangle $DPM$ has a right angle at $P$, with $PM = \frac{1}{2\sqrt{3}}$ and hypotenuse $DM = \frac{\sqrt{3}}{2}$. Hence

$$\cos\theta = \frac{PM}{DM} = \frac{1/(2\sqrt{3})}{\sqrt{3}/2} = \frac{1}{3},$$

as required. (As a check, $PD^2 = DM^2 - PM^2 = \frac{3}{4} - \frac{1}{12} = \frac{2}{3}$, agreeing with part (i).)

**Part (iii)**

**The best sphere.** Let $O$ be the point of $DP$ with $OP = r$, where $r$ is to be found so that $O$ is also at distance $r$ from the three sloping faces. By symmetry, it is enough to deal with the face $DBC$.

The plane $AMD$ is perpendicular to $BC$ (it contains $AM$ and $DM$, both perpendicular to $BC$), so it is perpendicular to the face $DBC$, and the perpendicular from $O$ to that face lies in this cross-section: it is the perpendicular from $O$ to the line $MD$. Since $O$ is at distance $OP$ from line $MP$ and at distance $r$ from line $MD$ and we want these equal, $O$ must lie on the bisector of angle $\theta$ at $M$. So

$$r = PM \tan\tfrac{\theta}{2}.$$

From part (ii), using $\tan^2\frac{\theta}{2} = \dfrac{1-\cos\theta}{1+\cos\theta}$,

$$\tan^2\tfrac{\theta}{2} = \frac{1-\frac{1}{3}}{1+\frac{1}{3}} = \frac{1}{2}, \qquad \tan\tfrac{\theta}{2} = \frac{1}{\sqrt{2}}$$

(positive, since $\theta$ is acute). Therefore

$$r = \frac{1}{2\sqrt{3}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2\sqrt{6}} = \frac{\sqrt{6}}{12}.$$

The sphere with centre $O$ and radius $\frac{\sqrt{6}}{12}$ touches all four faces, and lies inside the tetrahedron because $O$ is an interior point at distance at least $r$ from every face.

? **Ask yourself** How do we know no sphere, placed anywhere, can be bigger than this one?

**Nothing larger fits.** Take any point $X$ inside the tetrahedron, at distances $d_1, d_2, d_3, d_4$ from the four faces. Joining $X$ to the vertices splits the tetrahedron into four pyramids, one on each face, each with base area $\frac{\sqrt{3}}{4}$. Adding their volumes,

$$V = \tfrac{1}{3} \cdot \tfrac{\sqrt{3}}{4} (d_1 + d_2 + d_3 + d_4).$$

But taking $X = D$ (where three distances are 0 and one is $PD$) shows $V = \frac{1}{3} \cdot \frac{\sqrt{3}}{4} PD$. Hence for every interior point

$$d_1 + d_2 + d_3 + d_4 = PD = \frac{\sqrt{6}}{3}.$$

A sphere of radius $\rho$ centred at $X$ fits inside only if every $d_i \geq \rho$, so $4\rho \leq \frac{\sqrt{6}}{3}$, that is, $\rho \leq \frac{\sqrt{6}}{12}$. So the largest sphere has radius

$$r = \frac{\sqrt{6}}{12} = \tfrac{1}{4} PD,$$

with equality exactly when all four distances are equal (at $O$).

## Checks

- Coordinates: $A = (0,0,0)$, $B = (1,0,0)$, $C = (\frac{1}{2}, \frac{\sqrt{3}}{2}, 0)$, $D = (\frac{1}{2}, \frac{\sqrt{3}}{6}, \sqrt{2/3})$ give all six edges of length 1, and $PD \approx 0.8165$.
- The outward unit normals of any two faces have dot product $-\frac{1}{3}$, confirming $\cos\theta = \frac{1}{3}$ (about $70.5°$).
- The point $(\frac{1}{2}, \frac{\sqrt{3}}{6}, \frac{\sqrt{6}}{12})$ is at distance $0.20412\ldots = \frac{\sqrt{6}}{12}$ from all four faces. Random interior points always gave a distance sum of $0.8165\ldots$, and their smallest distance never exceeded 0.2041.
- The centre is a quarter of the way up $PD$. This matches the fact that the incentre of a regular tetrahedron is its centroid, which lies a quarter of the way from a face to the opposite vertex.

## Common slips

- Using $\angle DAM$ or $\angle DBA$ as the angle between faces. The lines must both be perpendicular to the common edge, which $DM$ and $AM$ are.
- Taking $PA$ to be half a median instead of two-thirds of one.
- Finding the sphere touching the four faces without saying why no sphere elsewhere can be larger. The distance-sum identity settles this in two lines.

## Beyond STEP

The constant sum of distances to the faces is the 3D version of Viviani's theorem for an equilateral triangle. It comes from barycentric coordinates, which describe any point of a simplex as a weighted average of its vertices. The angle $\arccos\frac{1}{3}$ appears in chemistry too: the angle between bonds in methane is $\arccos(-\frac{1}{3}) \approx 109.5^\circ$, the angle subtended at the centre by two vertices of this same tetrahedron. For more of this kind of 3D reasoning, see Siklos, *Advanced Problems in Mathematics*, or Strang, *Introduction to Linear Algebra* for the vector approach with dot and cross products.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.